
Absenteeism in the Workplace

PRESENTED BY

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Research question(s)

REGRESSION

Can we predict absenteeism
hours?



CLASSIFICATION

Can we categorize absence
frequency?

Models



Multiple Linear Regression

The diagram consists of four horizontal light blue rounded rectangles, each containing a model name. To the left of each rectangle is a short vertical line. From the bottom of each of these four lines, a horizontal line extends to the right, connecting to the left side of the next rectangle. This creates a series of four connected horizontal segments, with the first and last segments being longer than the middle two.

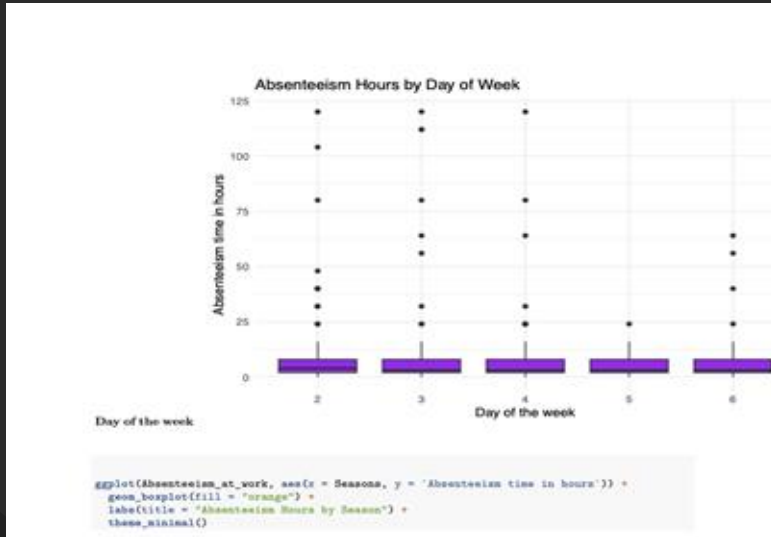
KNN

Logistic Regression

Tree

Absentee hours across groups

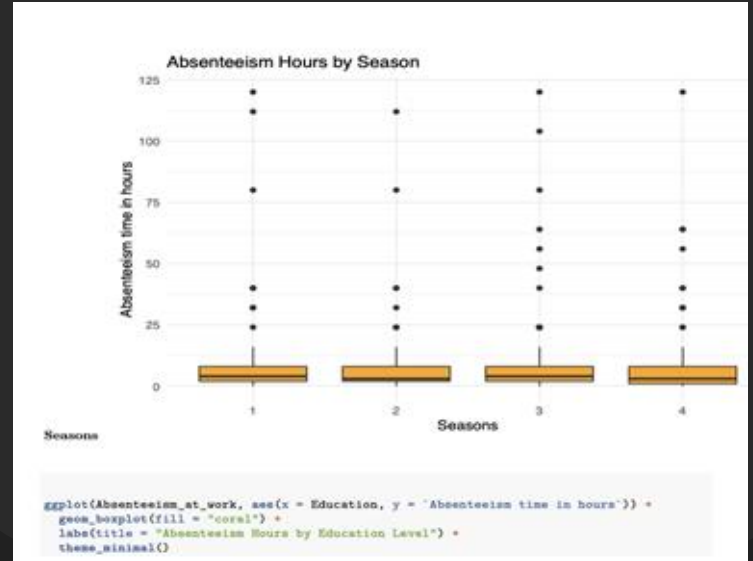
ABSENTEEISM BY DAY OF WEEK



KEY FINDING

Median absence is similar across days — day-of-week patterns look weak.

ABSENTEEISM BY SEASON

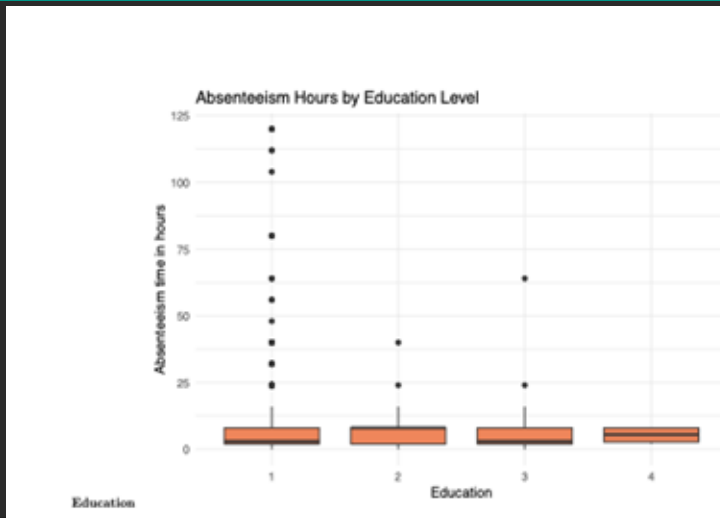


KEY FINDING

Seasonal differences are small — outliers appear in every season.

Education levels and absence

ABSENTEEISM BY EDUCATION LEVEL



KEY FINDING

Absence hours are low across all education levels — but Level 1 shows more extreme outliers.

Fitted MLR: setup and split

PREPROCESSING

```
## R
# Fitted Model (Regression): Multiple Linear Regression Baseline
set.seed(123)
library(tidyverse)

df <- Absenteeism_at_work

# convert categorical columns to factors (only if they exist)
cat_vars <- c("Reason for absence", "Month of absence",
              "Day of the week", "Seasons", "Education")
df <- df %>%
  mutate(across(any_of(cat_vars), as.factor))

# drop obvious non-features / target-derivative columns if present
drop_cols <- intersect(names(df), c("ID", "absence_category", "absence_binary"))
feat_cols <- setdiff(names(df), c("Absenteeism time in hours", drop_cols))

# 70/30 train-test split
idx <- sample(seq_len(nrow(df)), size = floor(0.7 * nrow(df)))
train <- df[idx, ]
test <- df[-idx, ]

# optional but safe: remove rows with missing values in each split
train <- train %>% drop_na()
test <- test %>% drop_na()
```

TRAIN / TEST SPLIT

```
# fit multiple linear regression
mlr_fit <- lm('Absenteeism time in hours' ~ .,
             data = train[, c("Absenteeism time in hours", feat_cols)])

# predict on hold-out test set
pred_mlr <- predict(mlr_fit, newdata = test)

# performance metrics
rmse_mlr <- sqrt(mean((pred_mlr - test$'Absenteeism time in hours')^2))
mae_mlr <- mean(abs(pred_mlr - test$'Absenteeism time in hours'))

rmse_mlr
mae_mlr
summary(mlr_fit)
```

SETUP STEPS

- Fit a multiple linear regression to predict absenteeism hours
- Converted reason, month, day, season, education into factors
- Dropped ID and outcome-duplicate columns
- 70/30 train-test split, removed rows with missing values

Multiple linear regression-model performance

0.23

ADJUSTED R²

13.4

TEST RMSE (HOURS)

7.2

TEST MAE (HOURS)

KEY FINDINGS

- **F-test is significant** — predictors matter jointly.
- **Adjusted R² $\approx$ 0.23** — fit is modest.
- **Test RMSE $\approx$ 13.4 hrs, MAE $\approx$ 7.2 hrs.**
- Statistically significant but only **moderately accurate**.

MODEL SUMMARY

```
lm(formula = Absenteeism.time.in.hours ~ ., data = train,
    ic("Absenteeism.time.in.hours", Fapt,col(323))

Residuals:
    Min       1Q   Median       3Q      Max
-34.288  -4.247  -0.689   2.478  35.826

Coefficients: (3 not defined because of singularities)
              (Intercept)              Estimate Std. Error t value Pr(>|t|)
1 (Intercept)          89.60884    93.13884    -0.744  0.454840
Reason for absence 1     6.34842     4.48655    1.379  0.171375
Reason for absence 2    23.88484    12.43474    1.919  0.062791 *
Reason for absence 3     9.49438    12.30111    0.772  0.440125
Reason for absence 4     4.29796     9.95884    0.474  0.635487
Reason for absence 5     9.76821     7.39928    1.319  0.187880
Reason for absence 6    26.85513     6.38809    4.193  0.000000 ***
Reason for absence 7    14.83517     4.68562    3.167  0.001643 **
Reason for absence 8    18.14427     7.61079    2.384  0.018811 *
Reason for absence 9    21.78817     6.86568    3.173  0.001428 **
Reason for absence 10   12.44266     8.85366    1.405  0.000770 ***
Reason for absence 11   11.09458     8.81775    1.259  0.021573 **
Reason for absence 12   38.25724     9.87487    3.869  0.000167 ***
Reason for absence 13   28.85744     3.14646    9.174  0.000000 ***
Reason for absence 14     9.35218     3.84638    2.431  0.015429 *
Reason for absence 15     9.53315    12.21734    0.782  0.434669
Reason for absence 16     6.32218     7.73499    0.817  0.414163
Reason for absence 17    11.12482    12.52443    0.888  0.374798
Reason for absence 18    15.28846     4.28996    3.561  0.000319 ***
Reason for absence 19   20.58423     3.58324    5.746  0.000000 ***
Reason for absence 20    13.57613     7.58789    1.789  0.071228 .
Reason for absence 21     9.60586     7.77428    1.249  0.099486 **
Reason for absence 22    3.50584     2.77937    1.257  0.051505 *
Reason for absence 23   18.81739     7.34498    2.572  0.011729 *
Reason for absence 24     6.87091     9.76461    0.703  0.479512
Reason for absence 25     7.92586     5.46896    1.449  0.022718 *
Reason for absence 26     8.88879     3.42577    2.592  0.009648 **
Reason for absence 27     5.27512     2.85073    1.851  0.064957 .
Month of absence 1     -1.83375     0.21246   -8.639  0.000000 ***
Month of absence 2     -0.72958     0.18456   -3.956  0.000000 ***
Month of absence 3     1.51146     0.56384    2.682  0.008343 **
Month of absence 4     -1.19534     0.89854   -1.331  0.099512
Month of absence 5     -5.08522     0.14471   -35.156  0.000000 ***
Month of absence 6     0.12623     0.86141    0.144  0.888881
Month of absence 7     0.36458     0.58323    0.625  0.531723
Month of absence 8     2.99577     0.60622    4.942  0.000000 ***
Month of absence 9     5.52651     0.68149    8.111  0.000000 ***
Month of absence 10    1.98778     0.79748    2.492  0.013897 *
Month of absence 11    3.64492     0.68183    5.346  0.000000 ***
Month of absence 12    4.03984     0.52673    7.671  0.000000 ***
Day of the week 1     -2.45776     0.81829   -2.991  0.002621 *
Day of the week 4     -2.22868     1.64384   -1.351  0.176543
Day of the week 5     -4.28518     1.78478   -2.399  0.017946 *
Day of the week 6     -4.21537     1.72039   -2.450  0.015651 *
Seasons2              2.96268     0.88967    3.331  0.000889 **
Seasons3              5.72358     3.58905    1.595  0.111341
Seasons4              2.37258     3.75882    0.631  0.524158
Transportation expense 0.82087     0.82173    1.000  0.017157 *
Distance From Residence to Work -0.89393     0.88879   -1.005  0.315706
Service time          0.87484     0.23248    3.762  0.000000 ***
Age                   0.49249     0.14312    3.434  0.000649 ***
Work Load Average/day 0.81831     0.49178    1.663  0.097157
Wk1 target           0.22996     0.23556    0.975  0.332947
Disciplinary Failure 1 NA NA NA
Education2            -2.90782     2.87961   -1.014  0.311262
Education3            -4.44813     2.28673   -1.945  0.052667 .
Education4            -0.58176     8.82489   -0.659  0.509485
Sex                   0.45495     0.56863    0.800  0.423876
Social drinker 1      -1.89944     1.84713   -1.028  0.302596
Social smoker 1       -1.48137     2.87478   -0.515  0.600376
Pet                   -0.56289     0.82055   -0.685  0.491246
Weight                -0.41483     0.57586   -0.718  0.473785
Height                0.20961     0.32237    0.651  0.004411
Body mass index       -0.22581     1.65287   -0.137  0.891437
***
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


```

Results and key takeaways for MLR

0.225

ADJUSTED R²

3.42

F-STATISTIC

< 0.001

P-VALUE

STRENGTHS

- Simple baseline, easy to interpret
- Shows linear effects clearly

LIMITATIONS

- Low adjusted R² — modest explanatory power
- Assumes linear / additive relationships

COMPARED TO KNN / TREE

Less flexible than non-linear methods and typically **less accurate** — but easier to explain to stakeholders.

COEFFICIENTS

'Month of absence' 7	5.96850	9.58323	0.623	0.533723
'Month of absence' 8	2.99577	9.69612	0.309	0.757489
'Month of absence' 9	5.52631	9.68149	0.571	0.568489
'Month of absence' 10	1.80780	9.79794	0.185	0.853697
'Month of absence' 11	1.64492	9.68183	0.170	0.865166
'Month of absence' 12	4.65904	9.52673	0.490	0.624679
'Day of the week' 3	-1.45776	1.76025	-0.828	0.408015
'Day of the week' 4	-2.22068	1.68304	-1.319	0.187683
'Day of the week' 5	-4.18358	1.78478	-2.299	0.021946 *
'Day of the week' 6	-4.21257	1.73599	-2.426	0.015653 *
Sessions2	2.96168	4.80567	0.725	0.468889
Sessions3	5.72558	3.50905	1.595	0.111341
Sessions4	2.37258	3.75802	0.631	0.528138
'Transportation expense'	0.80307	0.81173	2.392	0.017157 *
'Distance from Residence to Work'	-0.80393	0.86079	-0.647	0.517926
'Service time'	-0.87484	0.23340	-0.321	0.748636
Age	0.49149	0.14312	3.434	0.000649 ***
'Work load Average/day'	-0.81831	0.81795	-1.874	0.283337
'Hit target'	0.22098	0.25356	0.871	0.383947
'Disciplinary failure' 1	NA	NA	NA	NA
Education2	-2.98702	2.87961	-1.818	0.313052
Education3	-4.44813	2.28673	-1.945	0.052367
Education4	-0.18176	8.82489	-0.812	0.990885
Sex	0.49705	0.58863	0.844	0.398076
'Social drinker' 1	-1.88844	1.84713	-0.595	0.552356
'Social smoker' 1	-1.48157	2.87470	-0.515	0.605336
Pet	-0.92989	0.63555	-1.463	0.144126
Weight	-0.81683	0.57800	-0.828	0.377885
Height	0.20961	0.52237	0.401	0.688411
'Body mass index'	-0.22561	1.65207	-0.137	0.894437
...				
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1				

KNN-model setup

KNN MODEL SETUP CODE

```
()  
#Fitted Model Z: k-NN (stratified split, NZV-safe)  
set.seed(123)  
library(dplyr)  
library(class)  
  
df <- Absenteeism_at_work  
drop_cols <- intersect(names(df), c("ID", "absence_category", "Absenteeism time in hours"))  
feat_cols <- setdiff(names(df), c("absence_binary", drop_cols))  
  
# reuse split if it exists; else make a stratified 70/30  
if (!exists("train") || !exists("test")) {  
  idx <- df %>% group_by(absence_binary) %>%  
    group_map(~ sample(seq_len(nrow(.x)), size = floor(0.7 * nrow(.x)))) %>% unlist()  
  train <- df[idx, ]; test <- df[-idx, ]  
}  
  
# numeric design matrices (drop intercept)  
X_tr <- model.matrix(absence_binary ~ ., data = train[, c("absence_binary", feat_cols)])[, -1]  
X_te <- model.matrix(absence_binary ~ ., data = test[, c("absence_binary", feat_cols)])[, -1]  
y_tr <- train$absence_binary  
y_te <- test$absence_binary
```

KEY STEPS

- Built a KNN classifier to predict absence severity (High vs. Low)
- Split on median absenteeism hours — anything above = High
- Stratified 70/30 split to keep class balance
- Kept only numeric features — KNN needs distances

KNN – Data Scaling

SCALING CODE

```
# drop zero-variance columns based on TRAIN (avoids NaN on scale)
nzv <- apply(X_tr, 2, sd) > 0
X_tr <- X_tr[, nzv, drop = FALSE]
X_te <- X_te[, nzv, drop = FALSE]
```

KEY STEPS

- Removed zero-variance predictors from the training set so they wouldn't distort distances
- Scaled all numeric features using training-set means and SDs
- Applied the same scaling to the test set for comparability
- Avoids data leakage and gives KNN a fair distance metric

KNN - Tuning K parameter

SCALING CODE

```
# scale using TRAIN stats
X_tr_sc <- scale(X_tr)
X_te_sc <- scale(X_te,
                 center = attr(X_tr_sc, "scaled:center"),
                 scale = attr(X_tr_sc, "scaled:scale"))
```

K TUNING LOOP

```
# tune k over odd values, then final fit
ks <- seq(3, 31, 2)
accs <- sapply(ks, function(k) mean(class::knn(X_tr_sc, X_te_sc, y_tr, k = k) == y_te))
k_best <- ks[which.max(accs)]
pred_knn <- class::knn(X_tr_sc, X_te_sc, y_tr, k = k_best)
acc_knn <- mean(pred_knn == y_te)

# report
k_best
acc_knn
table(Predicted = pred_knn, Actual = y_te)
```

KEY STEPS

- Tested odd K values from 3 to 31
- Evaluated each K using accuracy on the hold-out test set

KNN - Performance Assessment

K = 27

OPTIMAL K

75%

TEST ACCURACY

KEY FINDING

Optimal **K = 27** reached **75% accuracy** on the test set.

87.6%

LOW RECALL

58.3%

HIGH RECALL

WHERE THE MODEL STRUGGLES

- Much better at identifying **Low absenteeism (87.6%)** than High (58.3%).
- Weaker High recall — KNN struggles with catching high-absence employees.

CONFUSION MATRIX

```
[1] 27
[1] 0.7506361
```

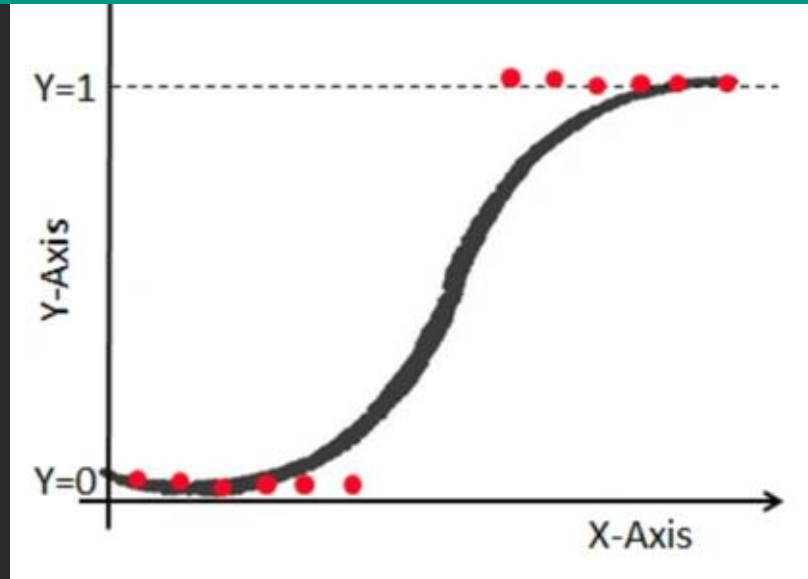
	Actual	
Predicted	High	Low
High	98	28
Low	70	197

Logistic

KEY FINDING

- Uses **median absence time** as the classifier
- The model performs **far better at predicting employees with high absenteeism**

LOGISTIC RESPONSE CURVE



Model Summary

MODEL SUMMARY

```
Call:
glm(formula = AbsentBinary ~ Age + Education + Service.time +
     Work.load.Average.day + Hit.target + Transportation.expense +
     Distance.from.Residence.to.Work + Body.mass.index, family = binomial,
     data = train)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.051e+00	2.716e+00	-0.387	0.699
Age	8.488e-04	2.127e-02	0.040	0.968
Education2	1.322e-01	4.062e-01	0.325	0.745
Education3	-5.271e-02	3.211e-01	-0.164	0.870
Education4	1.439e-01	1.431e+00	0.101	0.920
Service.time	1.130e-02	3.325e-02	0.340	0.734
Work.load.Average.day	2.700e-06	2.364e-06	1.142	0.253
Hit.target	-1.225e-02	2.425e-02	-0.505	0.613
Transportation.expense	7.133e-03	1.631e-03	4.373	1.23e-05 ***
Distance.from.Residence.to.Work	-8.196e-03	7.466e-03	-1.098	0.272
Body.mass.index	-7.419e-03	2.618e-02	-0.283	0.777

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

KEY FINDING

- Only **Transportation Expense** is statistically significant ($p < 0.001$)
- All other predictors, including **Age** and **Education**, show no significant effect

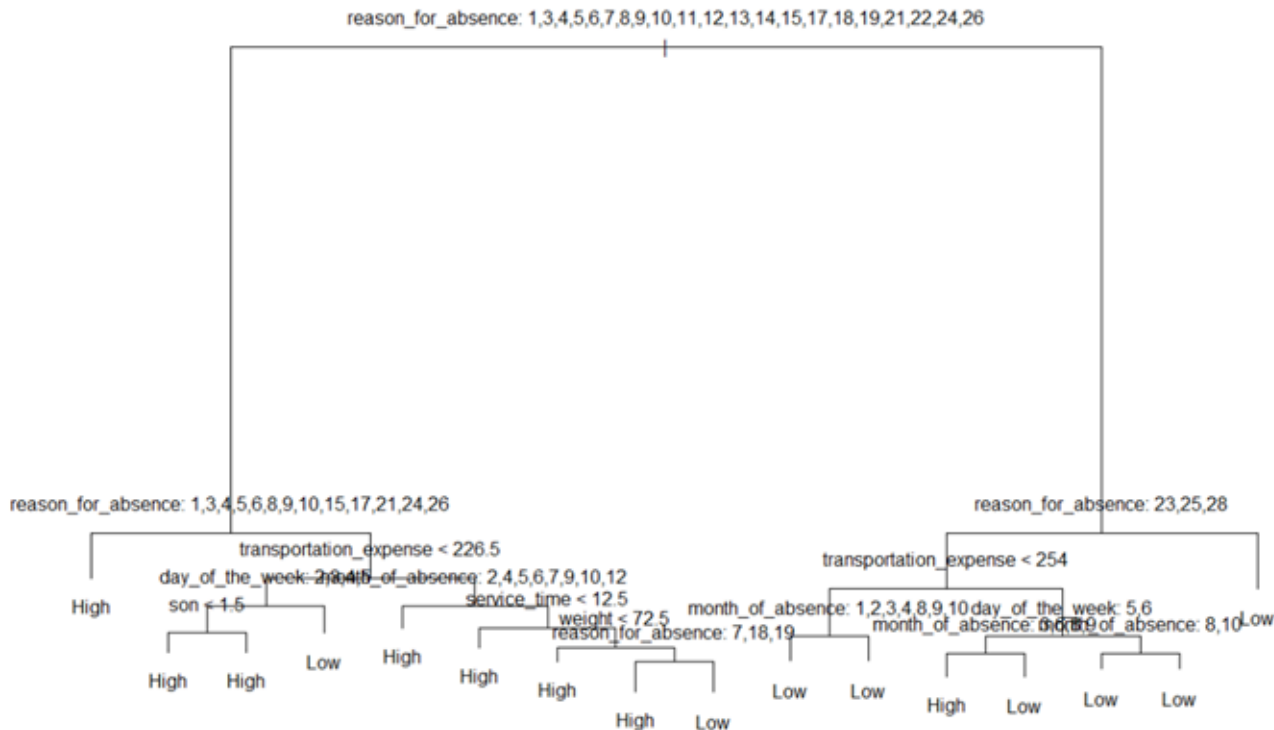
Tree

MODEL SETUP

Removed ID and two categories that were not independent from the response. Produced 16-node tree with residual mean deviance of 0.4544 and error rate of 0.1025.

```
library(tree)
library(janitor)
train <- train %>% clean_names()

absentee_tree <- tree(absence_binary ~ . - id - absenteeism_time_in_hours -
  absence_category, data=train)
```



Tree before pruning

Full (unpruned) tree: 16 terminal nodes, with the first split on transportation expense

```
node), split, n, deviance, yval, (yprob)
  = denotes terminal node

1) root 517 716.700 Low ( 0.49903 0.50097 )
  2) reason_for_absence: 1,3,4,5,6,7,8,9,10,11,12,13,14,15,17,18,19,21,22,24,26 246 197.600 High ( 0.86179 0.13821 )
    4) reason_for_absence: 1,3,4,5,6,8,9,10,15,17,21,24,26 70 0.000 High ( 1.00000 0.00000 ) =
    5) reason_for_absence: 7,11,12,13,14,18,19,22 176 172.800 High ( 0.80682 0.19318 )
      10) transportation_expense < 226.5 82 102.400 High ( 0.68293 0.31707 )
        20) day_of_the_week: 2,3,4,5 76 87.600 High ( 0.73684 0.26316 )
          40) son < 1.5 60 76.380 High ( 0.66667 0.33333 ) =
          41) son > 1.5 16 0.000 High ( 1.00000 0.00000 ) =
        21) day_of_the_week: 6 6 0.000 Low ( 0.00000 1.00000 ) =
      11) transportation_expense > 226.5 94 54.720 High ( 0.91489 0.08511 )
        22) month_of_absence: 2,4,5,6,7,9,10,12 49 0.000 High ( 1.00000 0.00000 ) =
        23) month_of_absence: 1,3,8,11 45 42.120 High ( 0.82222 0.17778 )
          46) service_time < 12.5 20 0.000 High ( 1.00000 0.00000 ) =
          47) service_time > 12.5 25 31.340 High ( 0.68000 0.32000 )
            94) weight < 72.5 8 0.000 High ( 1.00000 0.00000 ) =
            95) weight > 72.5 17 23.510 High ( 0.52941 0.47059 )
              190) reason_for_absence: 7,18,19 10 6.502 High ( 0.90000 0.10000 ) =
              191) reason_for_absence: 13,14,22 7 0.000 Low ( 0.00000 1.00000 ) =
        3) reason_for_absence: 0,23,25,27,28 271 246.900 Low ( 0.16974 0.83026 )
          6) reason_for_absence: 23,25,28 201 216.200 Low ( 0.22886 0.77114 )
            12) transportation_expense < 254 146 112.900 Low ( 0.13014 0.86986 )
              24) month_of_absence: 1,2,3,4,8,9,10 105 99.300 Low ( 0.18095 0.81905 ) =
              25) month_of_absence: 5,6,7,11,12 41 0.000 Low ( 0.00000 1.00000 ) =
            13) transportation_expense > 254 55 76.230 Low ( 0.49091 0.50909 )
              26) day_of_the_week: 5,6 36 47.090 High ( 0.63889 0.36111 )
                52) month_of_absence: 3,6,8,9 16 7.481 High ( 0.93750 0.06250 ) =
                53) month_of_absence: 2,4,7,10,11,12 20 26.920 Low ( 0.40000 0.60000 ) =
              27) day_of_the_week: 2,3,4 19 19.560 Low ( 0.21053 0.78947 )
                54) month_of_absence: 8,10 8 11.090 Low ( 0.50000 0.50000 ) =
                55) month_of_absence: 2,3,4,6,9,11 11 0.000 Low ( 0.00000 1.00000 ) =
            7) reason_for_absence: 0,27 70 0.000 Low ( 0.00000 1.00000 ) =
```

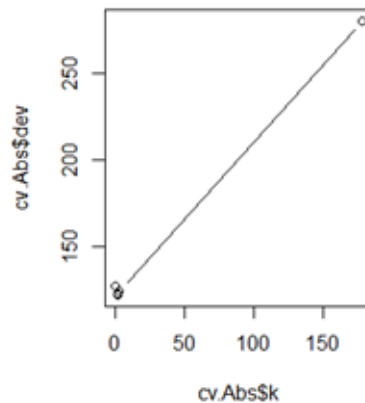
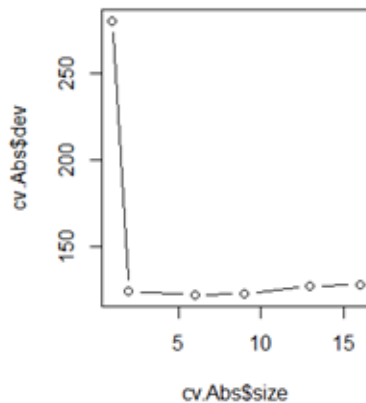

Tree cross-validation

CV CODE & OUTPUT

```
{r}  
set.seed(627)  
cv.Abs <- cv.tree(absentee_tree, FUN = prune.misclass, K=10)  
cv.Abs
```

```
$size  
[1] 16 13 9 6 2 1  
  
$dev  
[1] 128 127 123 122 124 280  
  
$k  
[1] -Inf 0.00 1.75 2.00 3.50 178.00  
  
$method  
[1] "misclass"  
  
attr(,"class")  
[1] "prune" "tree.sequence"
```

DEVIANE VS. TREE SIZE / K



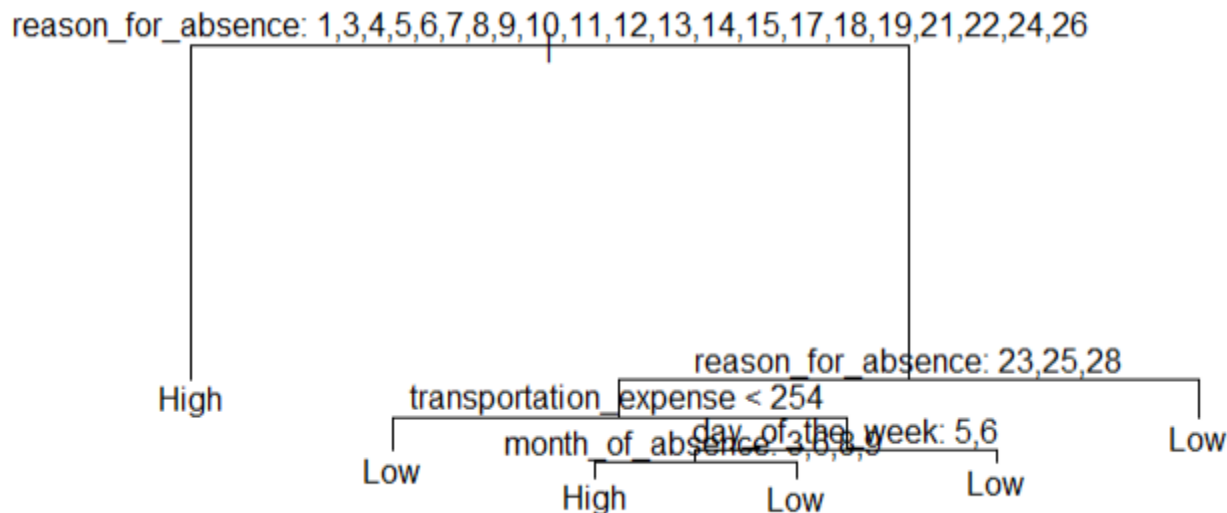
10-fold CV minimizes deviance at 6 nodes, supporting the prune-to-6 decision.

Tree pruning

PRUNING RESULTS

Pruning produced a 6-node tree with a residual mean deviance of 0.7133 and an error rate of 0.1277.

```
prune.Abs <- prune.misclass(absentee_tree, best = 6)
plot(prune.Abs)
text(prune.Abs, pretty = 0)
```



Model Comparison

MODEL COMPARISON METRICS

- The MLR uses RMSE, MAE, and R^2
- KNN, Logistic, and Decision Tree can all be compared with accuracy compared to testing data and confusion matrices

```
Residual standard error: 11.68 on 455 degrees of freedom  
Multiple R-squared:  0.318,    Adjusted R-squared:  0.225  
F-statistic: 3.421 on 62 and 455 DF,  p-value: 3.459e-14
```

Testing accuracy and matrices

KNN accuracy = 0.7506361

KNN confusion matrix:

Predicted	High	Low
High	98	28
Low	70	197

Logistic accuracy = 0.6471

Logistic confusion matrix:

Prediction	0	1
0	92	50
1	28	51

Pruned tree accuracy = 0.781

Pruned tree confusion matrix:

Confusion matrix on the TESTING data:

testtree.pred	High	Low
High	126	44
Low	42	181

Reflection/what we learned

KEY TAKEAWAYS

- Tree was the best at predicting frequency of absent workers
- With an accuracy rate of 78% for tree, vs KNN 75.1%, 64.7% for logistic
- MLR had adjusted R squared of 23%, and RMSE was off by 13.4 hours
- Worker health, age, and transportation expense were found to be some of the biggest reasons for absenteeism